PREMIER LEAGUE PREDICTOR

Table, points, and storyline in one view

Drop in your latest season snapshot or upload a raw match log. The FastAPI service loads the Random Forest stored at `src/Training/rf_model_2024_25.joblib`, recreates the medallion gold features, and sends back projected points plus positions.

Required fields per club: `team`, `wins`, `draws`, `losses`, `goalsFor`, `goalsAgainst`, `homeWins`, `awayWins`, and `isPromoted`. Matches played and points are filled in when missing.

Workflow

1. Edit the JSON payload so it reflects the table you want to score (20 rows for the full league).
2. Hit **Predict Table** or upload a CSV/XLS file if that's easier.
3. Review the response table and keep iterating. Nothing is stored server-side.

JSON playground

```
    "team": "Arsenal",
    "wins": 26,
    "draws": 6,
    "losses": 6,
    "goalsFor": 88,
    "goalsAgainst": 32,
    "homeWins": 14,
    "awayWins": 12,
    "isPromoted": 0
  },
  {
    "team": "Ipswich",
    "wins": 11,
    "draws": 9,
    "losses": 18,
    "goalsFor": 44,
    "goalsAgainst": 69,
    "homeWins": 7,
    "awayWins": 4,
```

Predict Table

Upload standings or match logs

Column names are case-insensitive. The app recognizes `team`, `wins`, `draws`, `losses`, `GF`, `GA`, `homeWins`, `awayWins`, and `isPromoted`. Uploading match-level files with `HomeTeam` / `AwayTeam` / `FTHG` / `FTAG` is also supported.

Choose File 2021-22.csv

Season 2024-25

Upload & Predict

Prediction output

PREDICTED POSITION	TEAM	SEASON	PREVIOUS POINTS	PREDICTED POINTS	PROMOTED
1	Man City	2024-25	93	80.56	No
2	Liverpool	2024-25	92	75.54	No
3	Arsenal	2024-25	69	68.95	No
4	Tottenham	2024-25	71	67.97	No
5	Man United	2024-25	58	61.46	No
6	Chelsea	2024-25	74	59.39	No
7	Brighton	2024-25	51	57.02	No
8	Aston Villa	2024-25	45	53.86	No
9	Newcastle	2024-25	49	53.46	No
10	Brentford	2024-25	46	53.04	No
11	Crystal Palace	2024-25	48	47.69	No
12	Wolves	2024-25	51	47.59	No
13	West Ham	2024-25	56	45.93	No
14	Leicester	2024-25	52	45.65	No
15	Burnley	2024-25	35	43.3	No
16	Everton	2024-25	39	39.1	No
17	Norwich	2024-25	22	37.81	No
18	Watford	2024-25	23	37.67	No
19	Southampton	2024-25	40	35.65	No
20	Leeds	2024-25	38	35.21	No

Feature columns under the hood

The app mirrors the gold dataset generated in Databricks, so FastAPI always feeds the model the same feature set.

prev_points prev_position prev_position_vs_avg prev_goal_difference prev_wins prev_draws prev_losses prev_points_per_game prev_win_pct prev_goals_per_game prev_goals_conceded_per_game prev_home_advantage prev_home_wins prev_away_wins is_promoted prev_position_category_numeric